

Vector Fields

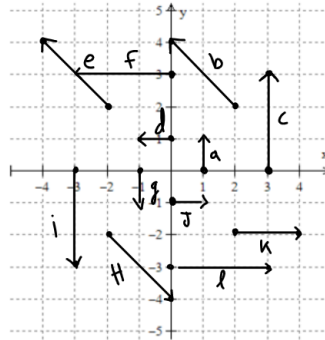
- a vector Field is like a map of arrows without a legend it describes air Flow, electric Fields, Magnetic Field and gravitational Field

★ the longer the arrow the stronger (larger in magnitude) the Vector is at that Point

- arrow direction → shows which way the Field Points
- arrow length → Shows how large or strong the Field is

3. Sketch the vector Field of $F(x, y) = -y\vec{i} + x\vec{j} = \langle -y, x \rangle$

	(x, y)	$F(x, y)$
a	(1, 0)	$\langle 0, 1 \rangle$
b	(2, 2)	$\langle -2, 2 \rangle$
c	(3, 0)	$\langle 0, 3 \rangle$
d	(0, 1)	$\langle -1, 0 \rangle$
e	(-2, 2)	$\langle -2, 2 \rangle$
f	(0, 3)	$\langle -3, 0 \rangle$
g	(-1, 0)	$\langle 0, -1 \rangle$
h	(-2, -2)	$\langle 2, -2 \rangle$
i	(-3, 0)	$\langle 0, -3 \rangle$
j	(0, -1)	$\langle 1, 0 \rangle$
k	(2, -2)	$\langle 2, 0 \rangle$
l	(0, -3)	$\langle 3, 0 \rangle$



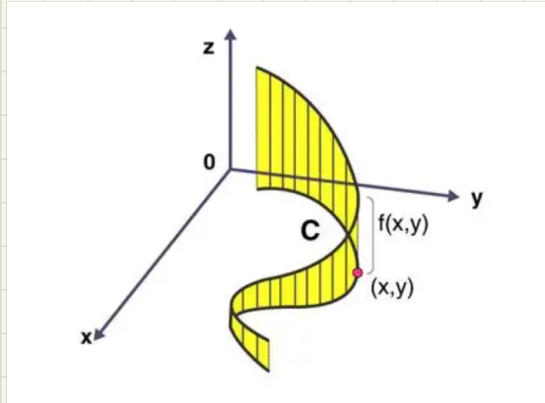
this describes where the Point is located

this describes the movement

line integral

So what exactly is a line integral?

a line integral is a type of integral where you integrate a function along a curve instead over an interval or over an area/volume



there are two main types and they "do" slightly different things depending on your integrating

line integrals of scalar fields

- these integrate a scalar-valued function (like temperature, density, or height) along a path

$$\int_C f(x, y) ds$$

line integrals of vector fields

- integrate a vector field (like force or velocity) along a curve

$$\int_C \vec{F} \cdot d\vec{r} \quad \cdot \text{taking dot product}$$

$$\int_C f(x,y) ds \Rightarrow \int_a^b f(x,y) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

you are replacing ds with $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$ then you are taking the derivative of the provided x and y or if there is a z component

examples:

evaluate the line integral $\int_C (x^2 + y^2 + z^2) ds$ over the space $C: x=t, y=\cos(4t), z=\sin(4t), 0 \leq t \leq 2\pi$

$$x=t \longrightarrow \frac{dx}{dt} = 1 dt$$

$$ds = \sqrt{(1)^2 + (-4\sin(4t))^2 + (4\cos(4t))^2}$$

$$y=\cos(4t) \longrightarrow \frac{dy}{dt} = -4\sin(4t)$$

$$ds = \sqrt{1 + 16\sin^2(4t) + 16\cos^2(4t)}$$

$$z=\sin(4t) \longrightarrow \frac{dz}{dt} = 4\cos(4t)$$

it doesn't matter what is in the $\sin$ and $\cos$ as long as they obey the identity

$$\sqrt{1 + 16(\sin^2(\theta) + \cos^2(\theta))}$$

$$\sqrt{1 + 16(1)} = \sqrt{17}$$

$$\int_0^{2\pi} (t^2 + \cos^2(4t) + \sin^2(4t)) \sqrt{17} dt$$

again this equals 1

$$\int_0^{2\pi} (t^2 + 1) \sqrt{17} dt \longrightarrow \sqrt{17} \int_0^{2\pi} t^2 + 1 dt \longrightarrow \left[\frac{t^3}{3} + t \right]_0^{2\pi}$$

$$= \left[\frac{(2\pi)^3}{3} + 2\pi \right] = \boxed{\frac{8\pi^3}{3} + 2\pi}$$

evaluate the line integral $\int_C y dx + z dy + x dz$, over the space curve $C: x = \sqrt{t}$

$$y = t, \quad z = t^2, \quad 1 \leq t \leq 9$$

$$x = \sqrt{t} \rightarrow \frac{dx}{dt} = \frac{1}{2\sqrt{t}} \rightarrow dx = \frac{1}{2\sqrt{t}} dt$$

$$y = t \rightarrow \frac{dy}{dt} = 1 \rightarrow dy = dt$$

$$z = t^2 \rightarrow \frac{dz}{dt} = 2t \rightarrow dz = 2t dt$$

$$\int_1^9 (t) \left(\frac{1}{2\sqrt{t}} dt \right) + (t^2) (dt) + (\sqrt{t}) (2t dt)$$

$$\int_1^9 \left(\frac{1}{2} \sqrt{t} + t^2 + 2t^{3/2} \right) dt \quad \cdot \text{Simplify}$$

$$\left[\frac{7t^{3/2}}{2} + \frac{t^3}{3} + \frac{4t^{5/2}}{5} \right]_1^9 \rightarrow \left[\left(\frac{7(9)^{3/2}}{2} + \frac{(9)^3}{3} + \frac{4(9)^{5/2}}{5} \right) - \left(\frac{7(1)^{3/2}}{2} + \frac{(1)^3}{3} + \frac{4(1)^{5/2}}{5} \right) \right]$$

$$\frac{5319}{10} - \frac{139}{30} = \boxed{527.27}$$

Find the work done by the force field $\vec{F}(x,y) = x\vec{i} + (y+1)\vec{j}$ in moving an objective an arch of cycloid $\vec{r}(t) = (t - \sin t)\vec{i} + (1 - \cos t)\vec{j}$, $0 \leq t \leq 2\pi$

$$\int_C \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt$$

$$\vec{r}(t) = \underbrace{(t - \sin t)}_{x(t)} \vec{i} + \underbrace{(1 - \cos t)}_{y(t)} \vec{j}$$

plug these into $\vec{F}$

$$\vec{F} = (t - \sin t) \vec{i} + (2 - \cos t) \vec{j}$$

take the derivative of $\vec{r}(t)$

$$\vec{r}'(t) = (1 - \cos t) \vec{i} + \sin t \vec{j}$$

take the dot product of $\vec{r}'(t)$ and $\vec{F}$

$$\vec{F} \cdot \vec{r}'(t) = (t - \sin t)(1 - \cos t) + (2 - \cos t) \sin t$$

Simplify

$$t - t \cos t - \sin t + \cancel{\cos t \sin t} + 2 \sin t - \cancel{\cos t \sin t}$$

$$t - t \cos t - \sin t + 2 \sin t$$

$$\int_0^{2\pi} (t + \sin t - t \cos t) dt$$

$$uv - \int v du \quad \text{liet}$$

$$u = t$$

$$dv = \cos t$$

$$du = dt$$

$$v = \int \cos t dt$$

$$v = \sin t$$

$$\left[t \sin t - \int \sin t dt \right] \Rightarrow t \sin t + \cos t$$

$$\left[\frac{t^2}{2} + \cos t + t \sin t + \cos t \right]_0^{2\pi}$$

$$\left(\left(\frac{(2\pi)^2}{2} + \cos(2\pi) + (2\pi) \sin(2\pi) + \cos(2\pi) \right) - \left(\cos(0) + \cos(0) \right) \right)$$

$$\boxed{2\pi^2}$$

Fundamental theorem of line integrals

$$\int_c \nabla f \cdot d\vec{r} = f(r(b)) - f(r(a))$$

if $a=b$ or the start is the same point as the end then:

$$\int_c \vec{F} \cdot d\vec{r} = 0$$

- Simply connected
- one piece, No holes
- Connected
- one piece, has hole



You will be given $\begin{cases} \vec{F}(x,y) = P\vec{i} + Q\vec{j} \\ \text{curve } c \end{cases}$

Find domain that is common to P and Q

Show that $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

Find the domain of $\partial P/\partial y$ and $\partial Q/\partial x$

Make sure all domains are simply connected which would make " $\vec{F}$ " conservative

if $\vec{F}$ is conservative then there exists a function f such that $\nabla f = \vec{F}$

now we can apply Fundamental theorem of line integral to find: $\int_c \vec{F} \cdot d\vec{r}$

1) $\vec{F}(x,y) = (y^2 - 2x)\vec{i} + 2xy\vec{j}$

P will always be i

Q will always be j

• if we have simply connected
its means we have conservative
vector field

- to see if we have simply connected:
- the domain is in one piece
- domain has no holes
- on that domain P, Q, P_y, Q_x must be defined (must have same domain as P, Q, P_y, Q_x)

if $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ and domain is in
one piece and domain has no
holes you can guarantee it
conservative

$P(x,y) = y^2 - 2x$
 $Q(x,y) = 2xy$ } • Find their domain for both which will be all real numbers so you can use the symbol $\mathbb{R}^2$

$\mathbb{R}$: you can plug any real number into the function the $\mathbb{R}^2$ means the function has two inputs a x and a y and you can plug in any number for both x and y

$$\frac{\partial P}{\partial y} = 2y$$

$$\frac{\partial Q}{\partial x} = 2y$$

• the $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ you next need to find the domain of both

domain of $\frac{\partial P}{\partial y} = \mathbb{R}^2$

domain of $\frac{\partial Q}{\partial x} = \mathbb{R}^2$

• the domain for P and Q is the same as P_y , Q_x also you have no holes in the domain which means you have a flat surface which translates to simply connected which means its conservative

all we did up to right now is show that its conservative we now need to find the actual "F" value which we will use this equation to find

$$\vec{F} = \frac{\partial F}{\partial x} \vec{i} + \frac{\partial F}{\partial y} \vec{j}$$

$$\frac{\partial F}{\partial x} = P \Rightarrow \frac{\partial F}{\partial x} = y^2 - 2x$$

$$\frac{\partial F}{\partial y} = Q \Rightarrow \frac{\partial F}{\partial y} = 2xy$$

• you will now use integration to find what F actually is

$$F = \int (y^2 - 2x) dx \quad \cdot \text{treat } y \text{ as a constant}$$

$$F = \int 2xy dy \quad \cdot \text{treat } x \text{ as a constant}$$

$$F = \int (y^2 - 2x) dx \Rightarrow F = xy^2 - x^2 + M$$

$$F = \int 2xy dy \Rightarrow F = xy^2 + m$$

• we integrated with respect to x
we lost y terms so we add m
was stands for missing

• we integrated with respect to y
so we lost x terms so we
add M which stands for
missing

With these two we can put the actual Function F together by
Finding something they both have in common in that that would be
 xy^2 and then you add all the extra stuff and add constant u

$$F = xy^2 - x^2 + u$$

For Conservative all that matters is where it starts and where it ends

$$2) F(x, y) = (y^2 \cos x + \cos y + 4) \vec{i} + (2y \sin x - x \sin y + e^y) \vec{j}$$

$$P(x, y) = y^2 \cos x + \cos y + 4$$

$$Q(x, y) = 2y \sin x - x \sin y + e^y$$

• the domain is all real number when you are
talking about $\cos$ or $\sin$ you can input
any value but $\cos'$ and $\sin'$ are restricted
between $[-1, 1]$ but we don't have an inverse
so you have domain as $\mathbb{R}^2$

$$\frac{\partial P}{\partial y} = 2y \cos x - \sin y \longrightarrow \text{domain of } \frac{\partial P}{\partial y} = \mathbb{R}^2$$

$$\frac{\partial Q}{\partial x} = 2y \cos x - \sin y \longrightarrow \text{domain of } \frac{\partial Q}{\partial x} = \mathbb{R}^2$$

So the $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ and the domain is the same the domain has no holes we can confirm this is simply connected and is conservative

For something to be conservative

1) $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

2) domain is simply connected meaning it has no holes this goes for domain of P and Q, and domain of $\frac{\partial P}{\partial y}$ and $\frac{\partial Q}{\partial x}$

$$\frac{\partial f}{\partial x} = P \text{ and } \frac{\partial f}{\partial y} = Q \longrightarrow \frac{\partial f}{\partial x} = y^2 \cos x + \cos y + 4 \text{ and } \frac{\partial f}{\partial y} = 2y \sin x - x \sin y + e^y$$

$$f = \int y^2 \cos x + \cos y + 4 \, dx \quad \text{and} \quad f = \int 2y \sin x - x \sin y + e^y \, dy$$

$$f = y^2 \sin x + x \sin y + 4x + m$$

$$f = y^2 \sin x + x \cos y + e^y + m$$

$$f = y^2 \sin x + x \sin y + 4x + e^y + m$$

they have $y^2 \sin x + x \cos y$ in common the rest is just the left over thats added up

$$\vec{F}(x, y, z) = yze^{xz} \vec{i} + e^{xz} \vec{j} + xye^{xz} \vec{k} \quad \text{and} \quad \vec{r}(t) = (t^2+1)\vec{i} + (t^2-1)\vec{j} + (t^2-2t)\vec{k}$$

$$0 \leq t \leq 2$$

a) Find a function such that $\vec{\nabla} f = \vec{F}$

$$\vec{r}(0) = \vec{i} - \vec{j} + 0\vec{k} = a(1, -1, 0)$$

$$\vec{r}(2) = 5\vec{i} + 3\vec{j} + 0\vec{k} = b(5, 3, 0)$$

~~✗~~ We assume its conservative

Find the function f such that $\vec{\nabla} f = \vec{F}$

$$\frac{\partial f}{\partial x} = yze^{xz}$$

$$\frac{\partial f}{\partial y} = e^{xz}$$

$$\frac{\partial f}{\partial z} = xye^{xz}$$

We take the integral to find function $\vec{F}$

$$f = \int yze^{xz} dx$$

$$f = \int e^{xz} dy$$

$$f = \int xye^{xz} dz$$

$$f = ye^{xz} + m$$

$$f = ye^{xz} + m$$

$$f = ye^{xz} + m$$

so your "F" function is:

$$f = ye^{xz} + k$$

Use Part a to evaluate the integral $\int_C \vec{F} \cdot d\vec{r}$

$$\int_C \vec{F} \cdot d\vec{r} = f(5, 3, 0) - f(1, -1, 0)$$

$$\left((3)e^{(5)(0)} + k \right) - \left((-1)e^{(1)(0)} + k \right)$$

$$= \boxed{4}$$

$\vec{F}(x,y) = (3+4xy^2)\vec{i} + (4x^2y)\vec{j}$ and C is an arc of a hyperbola $y = \frac{1}{x}$

From $(1,1)$ to $(3, \frac{1}{3})$

start

end

$$P(x,y) = 3 + 4xy^2$$

$$\frac{\partial P}{\partial y} = 8xy$$

$$Q(x,y) = 4x^2y$$

$$\frac{\partial Q}{\partial x} = 8xy$$

P and Q have a domain of all real numbers so $\mathbb{R}^2$

$\frac{\partial P}{\partial y}$ and $\frac{\partial Q}{\partial x}$ have a domain of all real numbers so $\mathbb{R}^2$

so these are conservative now you need to find the $F(x,y)$

$$\frac{\partial F}{\partial x} = P$$

$$\frac{\partial F}{\partial x} = 3 + 4xy^2$$

$$F = \int (3 + 4xy^2) dx$$

$$F = 3x + 2x^2y^2 + m$$

$$\frac{\partial F}{\partial y} = Q$$

$$\frac{\partial F}{\partial y} = 4x^2y$$

$$F = \int 4x^2y dy$$

$$F = 2x^2y^2 + m$$

$$F = 2x^2y^2 + 3x + n$$

this is what they both have in common the rest is just additional stuff

now use part a to evaluate the integral $\int_C \vec{F} \cdot d\vec{r}$

$$\int_C \vec{F} \cdot d\vec{r} = F(3, \frac{1}{3}) - F(1,1)$$

$$\left[2(3)(\frac{1}{3})^2 + 3(3) + n \right] - \left[2(1)(1)^2 + 3(1) + n \right]$$

$$\left(\frac{6}{9} + 9 + n \right) - (5)$$